ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PAPER 1 · PRACTICE E

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET first paper. Section A (30): multiple choice, true or false and matching · Section B (70): the motor effect and electromagnets, voltage dividers, efficiency, energy and cost, back emf, doping, LEDs and photodiodes, LED resistors, binary and hexadecimal, NOR gates and Boolean simplification, and desoldering. Formula sheet at the end.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B, and a formula sheet.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. Draw diagrams neatly, using a ruler. A non-programmable calculator may be used.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 The direction of the force on a current-carrying conductor in a magnetic field is found with …

A Lenz's law. B Ohm's law.
C Fleming's left-hand rule. D Kirchhoff's voltage law.

1.2 Adding a small amount of boron to pure silicon produces …

A P-type material. B N-type material.
C an insulator. D a superconductor.

1.3 The unit of magnetic flux density is the …

A weber. B henry. C farad. D tesla.

1.4 A machine that changes mechanical energy into electrical energy is a …

A motor. B generator. C transformer. D relay.

1.5 The efficiency of a machine is its output power …

A divided by its input power. B multiplied by its input power.
C plus its input power. D minus its input power.

1.6 Which diode gives off light when it is forward biased?

A Zener diode B LED
C Photodiode D Rectifier diode

1.7 The hexadecimal digit F is equal to the decimal number …

A 10. B 12. C 14. D 15.

1.8 Two equal resistors in series across 12 V form a voltage divider. The voltage across each resistor is …

A 12 V. B 24 V. C 6 V. D 3 V.

1.9 Which tool is used to remove solder from a joint?

A Desoldering pump
B Side cutters
C Crimping tool
D Wire stripper

1.10 In a PLC system, a proximity sensor is connected to the …

A output module.
B input module.
C power supply only.
D communication port.

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.10) in the ANSWER BOOK.

2.1 Like magnetic poles repel each other.

2.2 A reverse-biased photodiode conducts more current when light falls on it.

2.3 The total resistance of resistors in series is smaller than the smallest resistor.

2.4 An electromagnet loses most of its magnetism when the current is switched off.

2.5 Increasing the number of turns on a coil makes its magnetic field weaker.

2.6 Electricity bills show energy in kilowatt-hours.

2.7 Electrons are the majority charge carriers in N-type material.

2.8 A NOR gate works like an OR gate followed by a NOT gate.

2.9 The tip of a hot soldering iron should be cleaned with a dry metal file.

2.10 A transistor in saturation has a large voltage between its collector and emitter.

(10)

[10]

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–M) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	Rule that gives the direction of the force on a conductor in a motor	A	efficiency
3.2	Impurity added to a pure semiconductor	B	electromagnet
3.3	Light-sensitive diode used in reverse bias	C	dopant
3.4	Unit of energy used on electricity bills	D	kilowatt-hour
3.5	Output power divided by input power	E	NOR

3.6	Number system with base 16	F	Fleming's left-hand rule
3.7	Two resistors that give a fraction of the supply voltage	G	hexadecimal
3.8	Coil on an iron core that is a magnet only while current flows	H	photodiode
3.9	Tool that sucks up molten solder	I	voltage divider
3.10	Gate whose output is 0 if any input is 1	J	desoldering pump
		K	NAND
		L	Lenz's law
		M	binary

(10)

[10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 State Fleming's left-hand rule. (3)

4.2 A conductor 0,25 m long carries a current of 6 A at right angles to a magnetic field with a flux density of 0,5 T. Calculate the force on the conductor. (3)

4.3 Give FOUR ways to make an electromagnet stronger. (4)

4.4 Explain how a relay uses an electromagnet to switch a circuit. (3)

4.5 Give TWO uses of electromagnets in robots or automated machines. (2)

[15]

QUESTION 5

5.1 A 2 kΩ resistor (R_1) and a 3 kΩ resistor (R_2) form a voltage divider across a 10 V supply. Calculate the voltage across R_2. (3)

5.2 Calculate the current through the voltage divider in QUESTION 5.1. (2)

5.3 A motor takes 500 W from the supply and delivers 400 W of mechanical power. Calculate its efficiency and its power losses. (3)

5.4 A 2 kW heater runs for 3 hours. Calculate the energy used and the cost if electricity costs R2,80 per kWh. (4)

5.5 Explain what back emf is in a DC motor, and why a motor draws a very large current at the moment it starts. (4)

5.6 A 4 Ω and an 8 Ω resistor are connected in series across 12 V. Calculate the current and the voltage across each resistor. (4)

[20]

QUESTION 6

6.1 Explain how N-type and P-type semiconductor materials are made. (4)

6.2 Explain the difference between an LED and a photodiode, and give ONE use of each in a robot. (4)

6.3 Draw the symbol of EACH of the following: Zener diode; photodiode; PNP transistor. (3)

6.4 Explain why an LED needs a series resistor, and calculate the resistor needed for an LED (2 V, 20 mA) on a 9 V supply. (4)

[15]

QUESTION 7

7.1 Convert the following numbers:

7.1.1 Decimal 45 to binary (2)

7.1.2 Binary 11011 to decimal (2)

7.1.3 Hexadecimal 2F to decimal (2)

7.2 Draw the symbol and the truth table of a TWO-input NOR gate. (5)

7.3 Simplify the Boolean expression $Q = A + A \cdot B$ and name the law you used. (3)

7.4 Describe FOUR steps to remove and replace a faulty component on a printed circuit board. (4)

7.5 Give TWO causes of a dry (bad) solder joint. (2)

[20]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER

FORMULA SHEET

Force on a conductor: $F = B \times I \times l$	Ohm's law: $V = I \times R$
Voltage divider: $V_2 = V_s \times R_2 \div (R_1 + R_2)$	Series: $R_T = R_1 + R_2$; Parallel: $1 \div R_T = 1 \div R_1 + 1 \div R_2$
Efficiency: $\eta = P_{out} \div P_{in} \times 100\%$	Losses = $P_{in} - P_{out}$
Energy: $E = P \times t$ (kWh = kW × h)	Cost = energy × tariff
DC motor: $E_b = V - I_a R_a$	LED resistor: $R = (V_s - V_{LED}) \div I$